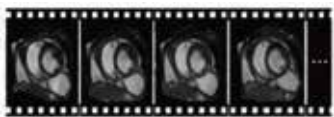


1 Heterogeneous multi-centre input

Multi-sequence stack



Cine film-strip



Missing sequences:
 $m = (m_{LGE}, m_{CO}, m_{T2})$

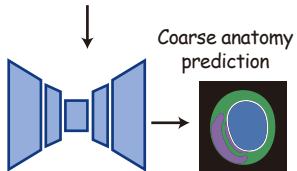
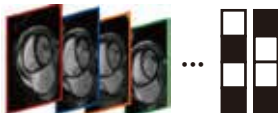


Missing labels:
 $s = (myo, LV, RV, edema, scar)$



2 Stage 1: anatomy conditioning

Input stack + presence maps



Shared anatomy prior

Spatial prior P



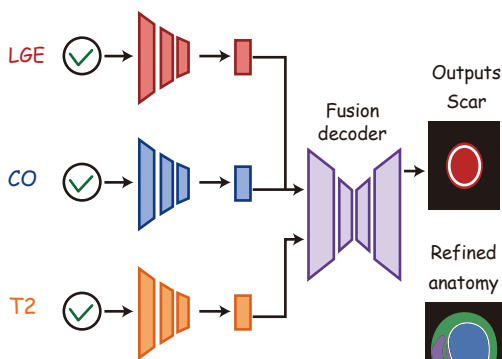
ROI Ω



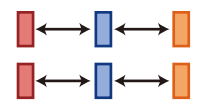
Conditioning prior (P / Ω)

3 Stage 2a: multi-sequence scar/anatomy expert

presence-gated encoders

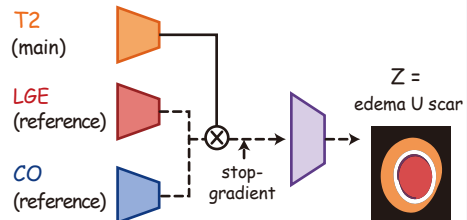


Optional feature alignment

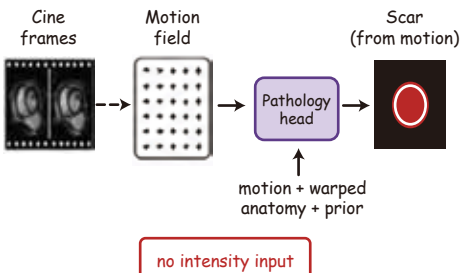


4 Stage 2b/2c: task-decoupled experts

Stage 2b: edema-zone expert



Stage 2c: cine motion scar expert



5 Label-level fusion & constrained output

Post-processing / inference

- flip TTA
- multi-scale cine
- anatomy containment
- small component removal

scar from Stage 2a



+

Z from edema expert

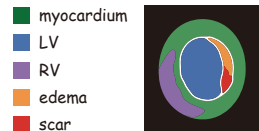


edema = Z \ scar

Constraint:

scar $\subset$ edema

Final five-label output



Sequence streams: LGE CO T2 cine

Arrow legend: data flow stop-gradient presence gate supervision mask

Misc legend: concat / features P / Ω conditioning prior